

HCWC Distribution Builder

From geological mechanisms to contact, risk and DHI — in one model.

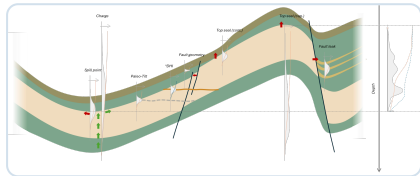
Build the HCWC distribution from competing geological limits.
Read the probability of success at every depth off the same sample.
Update with DHI evidence without discarding the geology.



1 The idea

Don't specify the HCWC — derive it from the geology.

- The tool models the **mechanisms that can stop a column**. Each defines its own limit.
- In every realisation all limits are sampled; the **shallowest active one** sets the contact.
- Across many realisations the distribution emerges from the competition — not from a shape somebody chose.



Every mechanism on one section, with the depth distribution at which it acts. Capacities are measured from the apex. Figure by Lars Hjelm.

LIMITS, COLOURED BY THE RISK ELEMENT THEY BELONG TO

- Fault geometry** · closure — juxtaposition across the bounding fault
- Capillary seal capacity** · retention — the column held before breakthrough
- Fault leakage** · retention — gouge, fault-rock, reactivation
- Reservoir pinch-out** · reservoir — connected pore volume ends
- Structural spill** · closure — what the trap geometry permits
- Charge limitation** · charge — not enough charge to fill deeper
- Seal continuity** · retention — a channel, window or thinning
- Mechanical failure** · retention — stress headroom, not entry pressure

2 How the tool works

1 Define the prospect

Apex, spill, the element chances giving $P(G)$, the assessment minimum, the well. Tab 2.0.

2 Elicit each limit

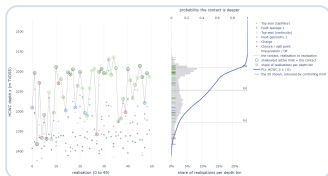
A capacity in metres below the crest, a mapped surface as a depth — whichever the geologist knows. Tab 3.0.

3 Run the competition

Each limit drawn for presence and depth; the shallowest active one sets the contact **and is recorded**. Tab 4.0.

4 Add DHI evidence — optional

The same realisations are reweighted. No second model, no arbitrary "DHI case". Tab 5.0.



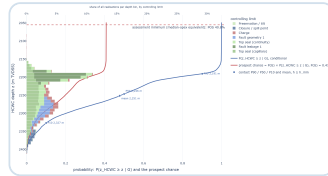
The competition, realisation by realisation: every limit's sampled depth, the shallowest ringed in its own colour, and the distribution those minima make.

$$H = \min\{h_i : \text{limit } i \text{ active}\}$$

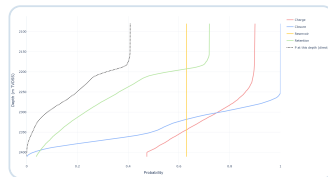
Minimum, not a blend — a limit only changes realisations that reach it. Blending a deep leak into a background distribution can make a prospect look *larger*.

3 Key outputs

- The **HCWC distribution**, and which mechanism controlled each realisation.
- Probability of success at every depth**, read at the assessment minimum and at the well.
- A curve per risk element**, for the volumetric tool downstream.
- All of it off **one sample**, so there is nothing to reconcile afterwards.



Blue is $P(z_{HCWC} \geq z | G)$; red is the prospect chance, $P(G)$ times it — the gap is the element risk. Bars are the controlling limit per depth bin.

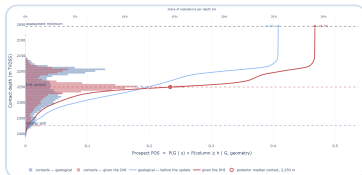


Each element's curve, derived from its own shallowest active limit rather than allocated by judgement.

4 DHI — update, don't replace

A seismic indication is evidence, not a separate case.

- The existing geological realisations are **reweighted**, not re-simulated.
- Each keeps its controlling limit, so prior and posterior are comparable figure for figure.
- How far the answer moves follows from **the strength of evidence stated**, not from a decision to believe the DHI.



The chance against threshold, geological and updated. The index *scales* the curve; the geometry *reshapes* it. The open circle is the posterior median, landing on the pick.

$$w_j \propto L(D | h_j, G)$$

Self-normalised importance sampling: the posterior is the same sample, reweighted.

What it is not. Substituting a DHI contact depth treats an observation as a scenario — and a scenario cannot be argued with on geological grounds.

5 Character vs geometry

One observation, two kinds of evidence — each updating exactly one factor.

CHANNEL 1 · CHARACTER

Are hydrocarbons present?

Amplitude, polarity, conformity, offset behaviour, placed on the evidence index. Updates the chance; **never moves the contact**.

$$LR(s) = f(s|HC) / f(s|NoHC)$$

CHANNEL 2 · GEOMETRY

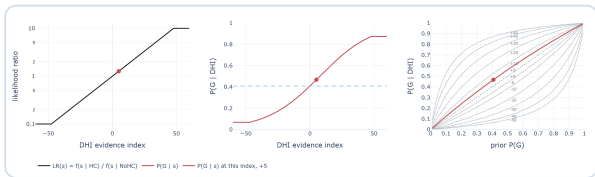
Where does the column end?

The picked depth and the confidence in it, as a likelihood over realisations already drawn. Reshapes the contact; **never moves $P(G)$** .

$$L = c \cdot D(h) \cdot \text{Pick}(z) + (1-c) \cdot s$$



The two reference distributions on the index. The likelihood ratio is the ratio of the two marked heights — which is why the axis units do not matter.

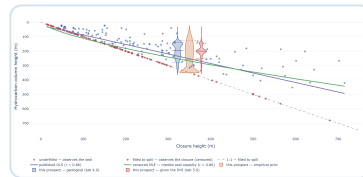


The ratio against the index, the posterior it gives, and the same update across priors — neutral evidence returning the prior on the diagonal.

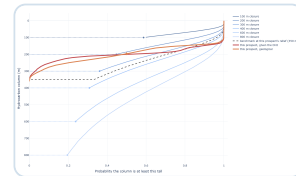
6 The empirical check

Filled-to-spill discoveries are right-censored, not measurements.

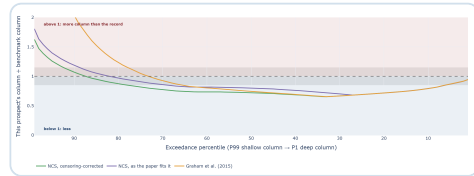
- Such a pool says the seal held **at least** the trap height — not what it could have held.
- Treated as exact values they bias a fit toward structural spill; **dropping them trades one bias for another**.
- The record is **compared, never joined**. A disagreement is a finding to explain, not a correction to apply.



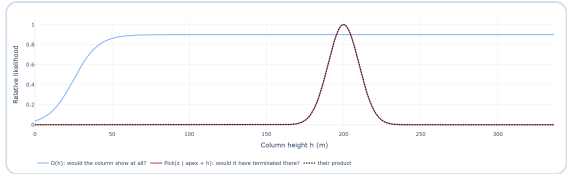
Red diamonds lie on the 1:1 line by definition — they record the closure, not the seal. The censored fit sits below the ordinary one because it estimates seal capacity.



Orange is the prospect; the dashed grey beside it is the benchmark at the same relief. Those two are the pair — the coloured curves are the family it sits inside, not its comparators.



The slope is the finding, not the level. A flat curve off parity is one correctable bias; a tilt is a disagreement about shape, which no single correction fixes.



$D(h)$ asks whether a column that size would have shown at all; the pick likelihood asks whether it would have terminated where the observed one did. Their product is the geometry evidence.

7 Keep these three apart

Merging them is where most of the confusion in this area comes from. Every probability the app prints says which it is, whether it includes $P(G)$, and the threshold it was read at.

QUANTITY	WHAT IT IS	CONDITIONAL ON
$P(G)$	An accumulation exists at all — the elements worked at the crest.	Nothing
$F(h) = P(H \geq h \mid G)$	The column reaches h , given an accumulation. What the limits produce.	An accumulation
$POS(h) = P(G) \times F(h)$	Hydrocarbons found at that depth — the product, and what an explorer wants.	Nothing

The commonest error. Quoting the conditional curve as the prospect's chance sets $P(G)$ to 1 and overstates the prospect by $1 / P(G)$ — largest exactly where the geology is least certain.

↓

Monotonic by construction
A deeper contact cannot be likelier than a shallower one — every point is read off one sample. A band-by-band array has no such guarantee.

≠

Uncertainty is not influence
A wide range can rarely decide the answer; a narrow one that usually decides it moves everything. Both are reported, separately.

ESS

Know what the answer rests on
Effective sample size says how concentrated the weighting became. A low value is not wrongness — it means the result leans on few realisations.

D

Absence is evidence, conditionally
A missing anomaly counts only in so far as that column would have been detectable. The ceiling on $D(h)$ is below 1 by construction.

8 The whole model, in one line

Risk elements
Charge, reservoir, closure, retention.

 $P(G) = \prod P_e$

Competing limits
Spill, charge, seal capacity and continuity, fault leakage, mechanical failure, pinch-out.

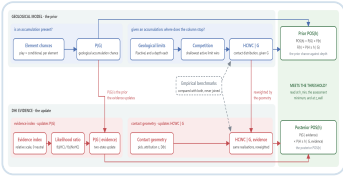
 $H = \min\{h_i\}$

HCWC distribution
Emerges from the competition, with the controlling mechanism recorded.

 $F(h) = P(H \geq h \mid G)$

Probability of success
The product, read at the minimum and at the well — then updated by evidence.

 $POS(h) = P(G) \cdot F(h)$



The geological model is the prior; the DHI is evidence. The index never moves the contact, the geometry never moves $P(G)$, each enters once, and one weighted sample reads every posterior quantity.

What it is not. Not a basin, hydrodynamic or 3D flow model. Where a more detailed model exists, its output enters as *one more competing limit*.

Notation. G the accumulation state · H column height, m · z_{apex} apex depth, m TVDSS · $z_{HCWC} = z_{apex} + H$ contact depth, increasing downward · h_{min} the assessment minimum · s the DHI evidence index · c the contact attribution. A probability written $| G$ is conditional on the accumulation.

Every figure on this sheet is the app's own output, exported by `scripts/export_exhibits.py` from the worked prospect that ships with the tool. The construction follows Beha, Christensen & Young (2012) and Hood (2019, 2024), run here as a continuous, correlated Monte Carlo; the per-element depth-dependent chance, the likelihood form of DHI evidence and the censoring correction are not in that literature. Derivations, worked numbers and references: tab 8.2. · Tool: [hcwc-builder.streamlit.app](#) · Source: [github.com/lhjelm-dk/HCWC-Builder](#) · Lars Hjelm